Chapter III

Section 1. Introduction and Motivation

Section 2. Series

Section 3. Convergent Series

Section 4. Series Tests

Series

1 Introduction and Motivation

One of the primary motivations for developing the subject of real analysis was precisely so that we could make sense of infinite series. In the opening chapters it was shown that treating infinite series as infinite summations led to paradoxical results and hinted that they must be treated with great care. It was also seen though that by treating infinite series as if they behaved algebraically like infinite summations, this could lead to practical results. This was seen in the example of integrating e^{-x^2} and assuming that integration of infinite series works the same as it does for finite summations. Despite this bold assumption, it led to an apparently correct and practical results. What we would like then is to build up a framework for infinite series that allows us to analyse what algebraic operations are permissible and which are not.

In the previous chapter, the topic of sequences was introduced through the realisation that any infinite series could define a sequence by considering finitely many terms of the series. Serious time was then spent building a framework to analyse the long term behaviour of sequences. In particular, it was seen that sequences could be either *convergent* or *divergent*. In the same way, a series can be either convergent or divergent depending on the nature of the sequence that is defined from a given series. Similar to sequences, a single real number can then be assigned to a series providing it it convergent. It is for this reason, that we will exclusively explore the properties of convergent series. This is mainly due to wanting to explore examples such as integrating e^{-x^2}. In order to produce the integral, we must first be able to equate a series to e^{-x^2} to proceed further. If we were to study divergent series, no value or function could be assigned to the series in the first place, so there is nothing to manipulate.
Despite this discussion, it is worth mentioning that divergent series are worth studying in their own right. They can yield important results and historically played a key role in the development of important convergent series, such as

$$\frac{1}{\pi} = \frac{2\sqrt{2}}{9801} \sum_{k=0}^{\infty} \frac{(4k)!(1103 + 26390k)}{(k!)^4 396^{4k}}.$$

The main reason for not studying them here is primarily due to their complicated nature. Much like the analysis of convergent series required great care, the analysis of divergent series requires even further care.

2 Series

The opening discussion outlined that every series can be assigned a sequence by considering finite portions of the series. This sequence could be convergent or divergent, depending on the given series. In this sense, the series is an object in

its own right and is defined independent of convergence. All we require then for a series to be defined is a sequence (a_k), from which a new sequence (s_n), with

$$s_n = a_1 + a_2 + \cdots + a_n,$$

can be defined. With this, we arrive at the following definition.

Definition 2.1: Series

Let (a_k) be a sequence. An ***infinite series***, or just ***series***, is a formal expression of the form

$$\sum_{k=1}^{\infty} a_k = a_1 + a_2 + \dots.$$

The ***sequence of partial sums*** (s_n) of a series is then defined by

$$s_n = a_1 + a_2 + \cdots + a_n$$

for each $n \in \mathbb{N}$.

It is important to realise that whilst the symbol $+$ and $\sum$ is used in the definition of an infinite series, these are certainly ***not*** the usual $+$ and summation found on $\mathbb{R}$. If they were, we could utilise properties like the commutativity of addition which we saw in the case of the infinite harmonic series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

led to contradictions. To add further confusion, the $+$ symbol used in the sequence of partial sums ***is*** the usual addition on $\mathbb{R}$. So whilst we use the same symbol for both, they actually mean very different things. This is similar to how $+$ is used in linear algebra to denote addition between matrices as well as element of $\mathbb{R}^n$. Despite the same notation, they act on different objects. In the case of real analysis, it is by moving from the finite to the infinite that the usual algebraic properties for addition on $\mathbb{R}$ break down. It is therefore best to think of $\sum_{k=1}^{\infty} a_k$ as simply an object in its own right not as a summation. A further important distinction to make is that the sequence (a_k) is distinct from the sequence (s_n). Do not get confused between these two sequences for their behaviour could be totally different.

3 Convergent Series

Given that any series can be assigned a sequence of partial sums, and any sequence can be either convergent or divergent, we can now provide a definition for a convergent/divergent series.

Definition 3.1: Convergent/Divergent Series

A series $\sum_{k=1}^{\infty} a_k$ is said to be ***convergent*** if the sequence of partial sums (s_n) converges. If (s_n) converges to $L \in \mathbb{R}$, the series is said to converge to L and is written as

$$\sum_{k=1}^{\infty} a_k = L.$$

The number $L \in \mathbb{R}$ is called the ***sum*** of the series.

If the sequence of partial sums (s_n) diverges, the series is said to be ***divergent.***

What this definition allows us to do is realise that in the special case of convergent series, we may assign a real number value to that series. So we initially think of series as abstract objects within their own right, and in the special case of convergent ones, they can simply be realised as real numbers.
One final comment to make is that since (a_k) and (s_n) are different, it is not the case that if (a_k) converges then (s_n) converges. Do not make the mistake of confusing these two sequences.
Given our definition of a convergent series, we can now utilise the framework of convergent sequences that was built up in the previous chapter to prove certain series are convergent/divergent.

Example 3.2: Convergent Series

The series given by $\sum_{k=1}^{\infty} 1/2^k$ converges to 1.

Proof. The sequence of partial sums (s_n) are given by

$$s_n = \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n}.$$

It follows by a proof by induction that

$$s_n = \frac{2^n - 1}{2^n}.$$

Hence, since $\lim(1/2^n) = 0$, it follows by the algebra of limits that $\lim(s_n) = 1$. □

This series was rather special in that we were able to construct a closed form expression for the sequence of partial sums. From this, we simply analysed the sequence using tools from the previous chapter. In fact, this type of proof would hold for an even wider class of series known as *geometric series*. This is outlined in the following theorem.

Theorem 3.3: Geometric Series

A series is called ***geometric*** if it is of the form

$$\sum_{k=0}^{\infty} ar^k,$$

with $r, a \in \mathbb{R} \setminus \{0\}$. Then a geometric series is convergent if and only if $|r| < 1$.

Proof. The sequence of partial sums (s_n) is given by

$$s_n = a + ar + \cdots + ar^n.$$

If $r = 1$, the series is clearly divergent. Otherwise, if $r \neq 1$, since

$$(1 - r)(a + ar + \cdots + ar^n) = a - ar^{n+1},$$

it follows that

$$s_n = \frac{a(1 - r^{n+1})}{1 - r}.$$

Hence, if $|r| < 1$, it follows that $\lim(r^{n+1}) = 0$ and thus by the algebraic limit theorem

$$\sum_{k=1}^{\infty} ar^k = \frac{a}{1 - r}.$$

Otherwise, if $|r| > 1$, it follows that the sequence (s_n) diverges and hence the series diverges. □

Whilst the tools developed in chapter 2 are useful for analysing sequences with closed form expressions, it is often the case with infinite series that the sequence of partial sums has no obvious closed form expression. Saying this, we can still utilise theorems on sequences such as the monotone convergence theorem and Cauchy criterion to analyse the behaviour of series, without knowing closed form expressions for s_n. Such an example of this is given here.

Example 3.4: Convergent Series

The series $\sum_{k=1}^{\infty} 1/k^2$ is convergent.

Proof. The sequence of partial sums (s_n) are given by

$$s_n = 1 + \frac{1}{4} + \cdots + \frac{1}{n^2}.$$

Since $0 \leq 1/k^2$ for each $k \in \mathbb{N}$, it follows that (s_n) is monotone increasing.

Now,

$$\begin{aligned} s_n &= 1 + \frac{1}{2\cdot 2} + \frac{1}{3\cdot 3} + \cdots + \frac{1}{n\cdot n} \\ &< 1 + \frac{1}{2\cdot 1} + \frac{1}{3\cdot 2} + \cdots + \frac{1}{n\cdot(n-1)} \\ &= 1 + \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right) \\ &= 1 + 1 - \frac{1}{n} \\ &< 2. \end{aligned}$$

Hence (s_n) is monotone increasing and bounded above so by the monotone convergence theorem, (s_n) is convergent. □

Much like with convergent sequence, we can construct convergent series from other convergent series. In particular, if any series is constructed out of the sum of convergent series, then it is convergent. Equally, if any series is constructed by a scalar multiple of a convergent series, it is also convergent. These statements are captured by the following theorem.

Theorem 3.5: Algebraic Limit Theorem for Series

Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be convergent series with $\sum_{k=1}^{\infty} a_k = \alpha$ and $\sum_{k=1}^{\infty} b_k = \beta$.

(i) $\sum_{k=1}^{\infty}(a_k + b_k) = \alpha + \beta$

(ii) $\sum_{k=1}^{\infty} ca_k = c\alpha$ for all $c \in \mathbb{R}$.

Proof. Exercise □

Notice here that unlike convergent sequences, there is no statement on the convergence of products or division of convergent series. The obvious answer to why there is no statement regarding this is because it isn't true. This is indeed the case but you should think about why this is the case. It is in fact that case that products of convergent series can be made convergent, but further restrictions are required and we don't have time to explore these series here.

4 Series Tests

As previously mentioned, not all series have nice closed form expressions for their sequence of partial sums. However, it is normally the case that the sequence (a_k) that defines the series will have a closed form expression. What would be nice then is if we could provide some tests that allow us to probe the sequence (a_k) and deduce some properties about the series $\sum_{k=1}^{\infty} a_k$. Indeed, since (a_k) and (s_n) are different sequences it maybe the case that our tests on (a_k) tell us nothing about the behaviour of (s_n), and this is sometimes the case. However, in special cases, it turns out that the behaviour of (a_k) tells us something about the behaviour of (a_k). The trick to realising the connection between the two sequences comes via

the Cauchy criterion on sequences. This is encapsulated by the following theorem.

Theorem 4.1: Cauchy Criterion for Series

A series $\sum_{k=1}^{\infty} a_k$ is convergent if and only if for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon$$

for every $n > m > N$.

Proof. The difference of the sequence of partial sums with $n > m$ is given by

$$s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n.$$

($\Rightarrow$) If (s_n) is convergent, it follows that it is Cauchy and hence for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|s_n - s_m| < \varepsilon$$

for every $n > m > N$.

($\Leftarrow$) If for every $\varepsilon > 0$ there exists a $N \in \mathbb{N}$ such that

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon$$

for every $n > m > N$ it follows that (s_n) is Cauchy. By the Cauchy criterion for sequences, it follows that (s_n) is convergent. □

So via the Cauchy criterion on sequences, we have managed to relate the convergence of the series to properties of the terms of the sequence (a_k). This immediately allows us to realise a necessary condition for the convergence of a series.

Theorem 4.2: Divergence Test

If the series $\sum_{k=1}^{\infty} a_k$ converges then $\lim(a_k) = 0$.

Proof. Letting $n = m + 1$ in Theorem 4.1, means that for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|a_{m+1} - 0| < \varepsilon$$

for every $m > N$. Hence $\lim(a_k) = 0$. □

By taking the contrapositive of the Theorem 4.2, we realise that if $\lim(a_k) \neq 0$, or $\lim(a_k)$ does not exist, it must be the case that the series $\sum_{k=1}^{\infty} a_k$ diverges. Hence why this test is aptly names the divergent test.

Example 4.3: Divergence Test

(i) Consider the series $\sum_{k=1}^{\infty}(-1)^k$. Here, $a_k = (-1)^k$. Since $\lim(a_k)$ does not exist, it follows by the divergence test that the series diverges.

(ii) Consider the series $\sum_{k=1}^{\infty}\frac{k}{k+1}$. Here $a_k = \frac{k}{k+1}$ and $\lim(k/(k+1)) = 1$. Hence, by the divergence test the series diverges.

(iii) Consider the series $\sum_{k=1}^{\infty}\frac{k}{k^2+1}$. Since $\lim(k/(k^2+1)) = 0$ nothing can be concluded via the divergence test about the nature of $\sum_{k=1}^{\infty}\frac{k}{k^2+1}$. That is, the series could be divergent or convergent, but the divergence test tells us nothing about which one it is.

The last example of the divergence test demonstrates the limitations of the divergence test. That is, if the sequence $(a_k) \to 0$, this tells us nothing about the nature of the series. Surely all we need to do then is prove the converse of the divergence test and all is sorted. Namely, if $(a_k) \to 0$, then the series $\sum_{k=1}^{\infty} a_k$ converges. Unfortunately, despite wishing this to be true, it is in fact false. A famous counterexample to this claim is given by the *harmonic series*.

Example 4.4: Harmonic Series

The series given by

$$\sum_{k=1}^{\infty}\frac{1}{k}$$

is called the ***harmonic series.*** The harmonic series is divergent.

Proof. The sequence of partial sums (s_{2^n}) is given by

$$\begin{aligned}
s_{2^n} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \cdots + \left(\frac{1}{2^{n-1}+1} + \cdots + \frac{1}{2^n}\right) \\
&> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}\right) + \cdots + \left(\frac{1}{2^n} + \cdots + \frac{1}{2^n}\right) \\
&= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2} \\
&= 1 + \frac{n}{2}.
\end{aligned}$$

Since (s_{2^n}) is unbounded, it follows that the series is divergent. □

Now, much like the proof of $\sum_{k=1}^{\infty} 1/2^k = 1$ led us to a general proof about geometric series, the proof of the divergence of the harmonic series leads us to a general statement about divergent series. In particular, since (a_k) with $a_k = 1/k$ is a monotone decreasing sequence, we can utilise the same techniques used to prove Example 4.4 to prove an equivalence between convergent/divergent series and series multiplied and indexed by powers of 2. This is otherwise known as the Cauchy condensation test.

Theorem 4.5: Cauchy Condensation Test

Let (a_k) be a monotone decreasing sequence with $a_k \geq 0$ for all $k \in \mathbb{N}$. Then, the series $\sum_{k=1}^{\infty} a_k$ converges if and only if

$$\sum_{k=0}^{\infty} 2^k a_{2^k}$$

converges.

Proof. Let (s_n) denote the sequence of partial sums for the series $\sum_{k=1}^{\infty} a_k$ and (t_m) denote the sequence of partial sums for the series $\sum_{k=0}^{\infty} 2^k a_{2^k}$. Then both (s_n) and (t_m) are monotone increasing since $a_k \geq 0$ for all $k \in \mathbb{N}$. For any $n \in \mathbb{N}$, let $m \in \mathbb{N}$ such that $2^m \leq n < 2^{m+1}$. It follows then that

$$\begin{aligned} s_n &= a_1 + a_2 + \cdots + a_n \\ &\leq a_1 + (a_2 + a_3) + \cdots + (a_{2^m} + \cdots + a_{2^{m+1}-1}) \\ &\leq a_1 + (a_2 + a_2) + \cdots + (a_{2^m} + \cdots + a_{2^m}) \\ &= a_1 + 2a_2 + \cdots + 2^m a_{2^m} \\ &= t_m. \end{aligned}$$

Since $2^m \leq n$ it follows that

$$\begin{aligned} t_m &= a_1 + 2a_2 + \cdots + 2^m a_{2^m} \\ &= 2(a_1/2 + a_2 + \cdots + 2^{m-1} a_{2^m}) \\ &\leq 2(a_1 + a_2 + \cdots + (a_{2^m} + \cdots + a_{2^m}) + a_{2^m+1} + \cdots + a_n) \\ &\leq 2(a_1 + a_2 + \cdots + a_n) \\ &= 2s_n. \end{aligned}$$

Hence, for every $n \in \mathbb{N}$, there exists an $m \in \mathbb{N}$ such that $s_n \leq t_m \leq 2s_n$. By the monotone convergence theorem, it follows that (s_n) is convergent if and only if (t_m) is convergent. □

At first glance, this theorem may seem a bit pointless. How can it be simpler to identify if a series if convergent by looking at the same series multiplied and indexed by powers of 2? If anything, surely this just makes the series more complicated. Well, it turns out that this test is extremely useful in identifying the behaviour of perhaps the next most simple series after geometric series, namely series of the form

$$\sum_{k=1}^{\infty} \frac{1}{k^p}$$

with $p \in \mathbb{R}$. We have already seen this series for when $p = 1$ and $p = 2$. For $p = 1$ the series was divergent whilst for $p = 2$ is was convergent. It is our goal then to deduce for what values of p this series is convergent. This is where the Cauchy condensation theorem comes to help since it related series of this type to geometric series, which we have already classified completely. This is shown explicitly in the following theorem.

Theorem 4.6: P-Series Test

The series

$$\sum_{k=1}^{\infty} \frac{1}{k^p}$$

is convergent if and only if $p > 1$.

Proof. By Theorem 4.5 it follows that $\sum_{k=1}^{\infty} 1/k^p$ is convergent if and only if

$$\sum_{k=0}^{\infty} 2^k \left(\frac{1}{2^k}\right)^p = \sum_{k=0}^{\infty} \left(\frac{1}{2^{p-1}}\right)^k.$$

Now, this is a geometric series with $r = 1/2^{p-1}$. By Theorem 3.3, the series is convergent if and only if $|1/2^{p-1}| < 1$. The proof is then complete since $|1/2^{p-1}| < 1$ if and only if $p > 1$. □

We have now built up a knowledge of a decent amount of convergent/divergent series. Namely, if a series is geometric or of the form $\sum_{k=1}^{\infty} 1/k^p$, for $p \in \mathbb{R}$, we can immediately say if it is convergent or not. What would be nice though is if we could compare new series to the series we know the behaviour of and deduce the behaviour of the initial series. This is precisely the point of the comparison test and is outlined in the theorem below.

Theorem 4.7: Comparison Test

Let (a_k) and (b_k) be sequences such that $0 \leq a_k \leq b_k$ for all $k \in \mathbb{N}$.

(i) If $\sum_{k=1}^{\infty} b_k$ converges, then $\sum_{k=1}^{\infty} a_k$ converges.

(ii) If $\sum_{k=1}^{\infty} a_k$ diverges, then $\sum_{k=1}^{\infty} b_k$ diverges.

Proof. (i) If $\sum_{k=1}^{\infty} b_k$ converges it follows by Theorem 4.1 that for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|b_{m+1} + b_{m+2} + \cdots + b_n| < \varepsilon$$

for every $n > m > N$. Since $0 \leq a_k \leq b_k$ for every $k \in \mathbb{N}$ it follows that

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon$$

for every $n > m > N$. Thus, by Theorem 4.1 the series $\sum_{k=1}^{\infty} a_k$ converges.

(ii) Follows from the contrapositive of (i). □

This test is aptly named the comparison test, since it tests the properties of series by comparing them to the known behaviour of other series. Note thought that the comparison test can only be applied if all the terms in the series are non-negative. This is the opening assumption of the theorem and therefore it cannot be utilised for series that have sequences (a_k) oscillating between positive and negative numbers. Some examples in which the comparison test can be utilised are given as

follows.

Example 4.8: Comparison Test

(i) Consider the series

$$\sum_{k=1}^{\infty} \frac{k+4}{k^3+1}.$$

Clearly

$$\frac{k+4}{k^3+1} \geq 0$$

for all $k \in \mathbb{N}$. Additionally,

$$\frac{k+4}{k^3+1} \leq \frac{k+4k}{k^3} = \frac{5}{k^2},$$

for all $k \in \mathbb{N}$. Since $\sum_{k=1}^{\infty} 1/k^2$ converges by the P-series test, and $\sum_{k=1}^{\infty} 5/k^2$ converges by Theorem 3.5, it follows by the comparison test that

$$\sum_{k=1}^{\infty} \frac{k+4}{k^3+1}$$

converges.

(ii) Consider the series

$$\sum_{k=1}^{\infty} \frac{k^2}{k^3+1}.$$

Then clearly

$$\frac{k^2}{k^3+1} \geq 0$$

for each $k \in \mathbb{N}$. Now,

$$\frac{k^2}{k^3+1} \geq \frac{k^2}{k^3+k^3} \geq \frac{1}{2k}$$

for all $k \in \mathbb{N}$. Since $\sum_{k=1}^{\infty} 1/k$ diverges, it follows that $\sum_{k=1}^{\infty} 1/(2k)$ diverges (this will be an exercise to prove), and hence by the comparison test

$$\sum_{k=1}^{\infty} \frac{k^2}{k^3+1}$$

diverges.

(iii) Consider the series

$$\sum_{k=1}^{\infty} \frac{(k+1)\sin^2(k)}{k^3+3k+1}.$$

Now, clearly

$$\frac{(k+1)\sin^2(k)}{k^3+3k+1} \geq 0$$

for each $k \in \mathbb{N}$. Additionally,

$$\frac{(k+1)\sin^2(k)}{k^3+3k+1} \leq \frac{k+k}{k^3} = \frac{2}{k^2},$$

for all $k \in \mathbb{N}$. By the P-series test and Theorem 3.5 $\sum_{k=1}^{\infty} 2/k^2$ converges, and by hence by the comparison test

$$\sum_{k=1}^{\infty} \frac{(k+1)\sin^2(k)}{k^3+3k+1}.$$

converges.

It was commented above that the comparison test can only be utilised if the individual terms within the sequence (a_k), defined from a series, are non-negative. Instead, is there a way to analyse the properties of series in which the sequence is of the form $(-1)^k a_k$? This is exactly the point of the *alternating series test* and will allow us to concretely say whether a series such as

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

converges.

Theorem 4.9: Alternating Series Test

Let (a_k) be a sequence such that

(i) (a_k) is monotone decreasing and

(ii) $\lim(a_k) = 0$.

Then, the alternating series $\sum_{k=1}^{\infty}(-1)^k a_k$ converges.

Proof. For $n > m$ it follows that

$$|s_n - s_m| = |a_{m+1} - a_{m+2} + a_{m+3} - a_{m+4} + \cdots \pm a_n|.$$

Since (a_k) is monotonically decreasing and $\lim(a_k) = 0$ it follows that $a_k \geq 0$ for each $k \in \mathbb{N}$. Hence, by induction it follows that

$$|s_n - s_m| \leq |a_{m+1}|$$

for all $n > m$. As $\lim(a_n) = 0$, it follows that for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $|a_m| < \varepsilon$ for every $m > N$. Thus, for every $n > m > N$, it follows that

$$|s_n - s_m| < \varepsilon.$$

Hence (s_n) is Cauchy and thus $\sum_{k=1}^{\infty}(-1)^k a_k$ converges. □

Again, it is worth commenting what this theorem tells us about alternating series and what it does not. Specifically, if we are given an alternating series of the form

$$\sum_{k=1}^{\infty}(-1)^k a_k$$

that has $\lim(a_k) = 0$ and (a_k) is not monotone decreasing, then the alternating se-

ries test tells us nothing about the convergence or divergence of $\sum_{k=1}^{\infty}(-1)^k a_k$. The alternating series test is only a sufficient condition for convergence of $\sum_{k=1}^{\infty}(-1)^k a_k$, but not a necessary one. The following example demonstrate how it is utilised.

Example 4.10: Alternating Series Test

(i) Consider the series

$$\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k}.$$

Then the sequence (a_k) is given by $a_k = 1/k$. Clearly $\lim(1/k) = 0$ and as

$$\frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)} \geq 0$$

for each $k \in \mathbb{N}$, it follows that (a_k) is monotone decreasing. Thus, by the alternating series test, and Theorem 3.5, it follows that

$$\sum_{k=1}^{\infty}(-1)\frac{(-1)^k}{k} = \sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k}$$

converges.

(ii) Consider the series

$$\sum_{k=1}^{\infty}\frac{(-1)^k}{\sqrt{k+1}}.$$

Then the sequence (a_k) is given by $a_k = 1/\sqrt{1+k}$. Again, it is clear that $\lim(1/\sqrt{1+k}) = 0$. Additionally,

$$\frac{1}{\sqrt{k+1}} - \frac{1}{\sqrt{k+2}} \geq \sqrt{k+2} - \sqrt{k+1} \geq 0,$$

since (b_n) with $b_n = \sqrt{n}$ is a monotone increasing sequence. Hence, by the alternating series test it follows that

$$\sum_{k=1}^{\infty}\frac{(-1)^k}{\sqrt{k+1}}$$

is convergent.

It may have been observed that all the series tests have so far relied on the individual terms of the sequence (a_k) being non-negative, or alternating in a very restrictive way. This is obviously not always the case as (a_k) could simply have some negative terms and some positive terms and not by alternating. How can we then analyse series that have this property using the tests we have developed? The obvious answer to this is we simply make all the terms positive using the absolute value function and use the series tests on these new series. This leads us naturally to the notion of absolute convergence.

Definition 4.11: Absolute Convergence

A series $\sum_{k=1}^{\infty} a_k$ is said to be ***absolutely convergent*** if $\sum_{k=1}^{\infty} |a_k|$ converges.

Now the notion of absolute convergence is only going to be of any use if an absolute convergent series is somehow related to the original series. In particular, to be able to apply any of the series tests developed we require that if a series is absolutely convergent, then it is convergent. If this is not the case, sticking absolute value functions on the terms of the sequence (a_k) would be about as useful as a chocolate teapot. Thankfully, this is not the case and is demonstrated in the following theorem.

Theorem 4.12: Absolute Convergence

If a series $\sum_{k=1}^{\infty} a_k$ converges absolutely, then it converges.

Proof. Suppose $\sum_{k=1}^{\infty} |a_k|$ converges. Since

$$0 \leq a_k + |a_k| \leq 2|a_k|$$

for each $k \in \mathbb{N}$, it follows by Theorem 4.7 and Theorem 3.5 that

$$\sum_{k=1}^{\infty} a_k + |a_k|$$

is convergent. Now, since $\sum_{k=1}^{\infty} |a_k|$ and $\sum_{k=1}^{\infty} a_k + |a_k|$ is convergent, it follows again by Theorem 3.5 that

$$\sum_{k=1}^{\infty} (a_k + |a_k|) - |a_k| = \sum_{k=1}^{\infty} a_k$$

is convergent. □

This theorem can then be utilised to extend the applicability of the comparison test by comparing the series with absolute value functions to a convergent series. This is captured in the following corollary.

Corollary 4.13: Comparison Test

Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be series with $b_k \geq 0$ for each $k \in \mathbb{N}$. Then if $0 \leq |a_k| \leq b_k$ for each $k \in \mathbb{N}$ and $\sum_{k=1}^{\infty} b_k$ is convergent, then $\sum_{k=1}^{\infty} a_k$ is convergent.

Without further ado, let's analyse some series using the properties of absolute convergence.

Example 4.14: Absolute Convergence

(i) Consider the series

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}.$$

Since the series

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^k}{k^2} \right| = \sum_{k=1}^{\infty} \frac{1}{k^2},$$

and $\sum_{k=1}^{\infty} 1/k^2$ is convergent by the P-series test, it follows that

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}.$$

is absolutely convergent.

(ii) Consider the series

$$\sum_{k=1}^{\infty} \frac{k \sin(k)}{k^3 + 5k^2 + 3}.$$

The sequence (a_k) is given by

$$a_k = \frac{k \sin(k)}{k^3 + 5k^2 + 3},$$

It follows then that

$$\begin{aligned} |a_k| &= \left| \frac{k \sin(k)}{k^3 + 5k^2 + 3} \right| \\ &\leq \frac{k}{k^3 + 5k^2 + 3} \\ &\leq \frac{k}{k^3} \\ &\leq \frac{1}{k^2}. \end{aligned}$$

Hence, by the P-series test $\sum_{k=1}^{\infty} 1/k^2$ converges and by Corollary 4.13 it follows that

$$\sum_{k=1}^{\infty} \frac{k \sin(k)}{k^3 + 5k^2 + 3}.$$

converges.

It is worth commenting now that it is certainly not the case that the converse of Theorem 4.12 is true. That is, a series is convergent implies it is absolutely convergent. One simple example of this is the series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}.$$

This is convergent, as shown in Example 4.10, but the series

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1}}{k} \right| = \sum_{k=1}^{\infty} \frac{1}{k}$$

is divergent, as shown in Example 4.4. Series that exhibit this property are given

a special name. That is, they are *conditionally convergent.*.

Definition 4.15: Conditional Convergence

A series $\sum_{k=1}^{\infty} a_k$ that does not converge absolutely, but still converges, it is said to be ***conditionally convergent.***

We conclude this chapter with one final test that relates the limit of the ratio of subsequent terms of the sequence (a_k) to the properties of the series $\sum_{k=1}^{\infty} a_k$. This test essentially is a combination of the absolute convergence test and comparison test, but comparing with geometric series. Intuitively, if the ratios of subsequent terms of the sequence (a_k) are less than 1, this means that sequence of partial sums of the series will eventually be dominated by the sequence of partial sums of a geometric series, with common ratio less than 1. Hence it must be convergent. If instead the ratios of subsequent terms of the sequence (a_k) are greater than 1, the sequence of partial sums of the series will eventual dominate the sequence of partial sums of a geometric series, with common ratio greater that 1. Hence, by the comparison test, it will be divergent. This intuitive idea is outlined in the following theorem.

Theorem 4.16: Ratio Test

Given a series $\sum_{k=1}^{\infty} a_k$, with $a_k \neq 0$ for any $k \in \mathbb{N}$, and

$$\lim \left| \frac{a_{k+1}}{a_k} \right| = L,$$

it follows that:

(i) if $L < 1$, the series converges absolutely.

(ii) if $L > 1$, the series is divergent.

(iii) if $L = 1$, the test is inconclusive.

Proof. (i) Suppose $L < 1$. Then there exists a $\alpha \in \mathbb{R}$ such that $0 \leq L < \alpha < 1$. Now, there exists an $N \in \mathbb{N}$ such that

$$\left| \left| \frac{a_{k+1}}{a_k} \right| - L \right| < \alpha - L$$

$$\Rightarrow \left| \frac{a_{k+1}}{a_k} \right| < \alpha,$$

for every $k > N$. It follows that

$$|a_{N+1+k}| < \alpha^k |a_{N+1}|$$

for every $k \in \mathbb{N}$. Since $\alpha < 1$, the geometric series given by $\sum_{k=1}^{\infty} \alpha^k |a_{N+1}|$ converges. By Theorem 4.7, $\sum_{k=N+2}^{\infty} |a_k|$ converges and hence $\sum_{k=1}^{\infty} |a_k|$ converges.

(ii) Suppose $L > 1$. Then there exist an $\alpha \in \mathbb{R}$ such that $1 < \alpha < L$. So, there exists an $N \in \mathbb{N}$ such that

$$\left|\left|\frac{a_{k+1}}{a_k}\right| - L\right| < L - \alpha$$

$$\Rightarrow \alpha < \left|\frac{a_{k+1}}{a_k}\right|,$$

for every $k > N$. Hence,

$$\alpha^k |a_{N+1}| < |a_{N+1+k}|$$

for each $k \in \mathbb{N}$. As $\alpha > 1$, the geometric series $\sum_{k=1}^{\infty} \alpha^k |a_{N+1}|$ is divergent and hence by Theorem 4.7 the series $\sum_{k=N+2}^{\infty} |a_k|$ diverges. Thus $\sum_{k=1}^{\infty} |a_k|$ diverges.

(iii) This is a tautology.

□

The use of this theorem is demonstrated in the following examples.

Example 4.17: Ratio Test

(i) Consider the series

$$\sum_{k=1}^{\infty} \frac{k}{5^k}.$$

Here, the sequence (a_k) is given by $a_k = k/5^k$. It follows then that

$$\left|\frac{k+1}{5^{k+1}} \frac{5^k}{k}\right| = \frac{k+1}{5k} \to \frac{1}{5}.$$

As $1/5 < 1$, it follows by the ratio test that

$$\sum_{k=1}^{\infty} \frac{k}{5^k}$$

converges.

(ii) Consider the series

$$\sum_{k=1}^{\infty} \frac{(k!)^2}{(2k)!}.$$

Then the sequence (a_k) is given by $a_k = \frac{(k!)^2}{(2k!)}$. It follows that

$$\left|\frac{((k+1)!)^2}{(2(k+1))!} \frac{(2k)!}{(k!)^2}\right| = \frac{(k+1)^2}{(2k+1)(2k+2)} \to \frac{1}{4}.$$

Since $1/4 < 1$, it follows by the ratio test that

$$\sum_{k=1}^{\infty} \frac{(k!)^2}{(2k)!}$$

converges.

(iii) Consider the series

$$\sum_{k=1}^{\infty} (-1)^{k+1} \frac{3^k}{k!}.$$

The sequence (a_k) is given by $a_k = (-1)^{k+1} 3^k / k!$. Then,

$$\left| \frac{(-1)^{k+2}}{(-1)^{k+1}} \frac{3^{k+1}}{(k+1)!} \frac{k!}{3^k} \right| = \frac{3}{k+1} \to 0.$$

As $0 < 1$, it follows by the ratio test that the series

$$\sum_{k=1}^{\infty} (-1)^{k+1} \frac{3^k}{k!}$$

converges.

This concludes the chapter on series. That isn't to say we wont see any more series. In fact, one of the main reasons for studying series was not to look at series of the form $\sum_{k=1}^{\infty} a_k$ where each $a_k \in \mathbb{R}$, but to make sense of series of the form $\sum_{k=1} f_k(x)$ where $f_k : \mathbb{R} \to \mathbb{R}$ is a real function for each $k \in \mathbb{N}$. In order to do this though, we must first analyse what it means for something to be a function and what properties that function may satisfy when defined on the real numbers. This is the point of the next chapter.